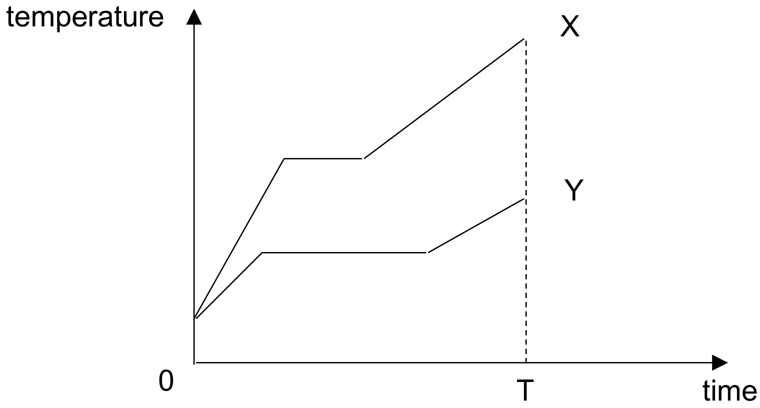
D is correct as **all** bodies are in thermal equilibrium.

15	Equal masses of two solids X and Y are heated from the same temperature. Energy is supplied by heating at the same constant rate to each solid.

[Turn over

The graph shows how the temperature of each solid varies with time after heating starts.



What can be deduced from the graph?

A	X has the higher boiling point.
B	X has the larger specific latent heat of fusion.
C	When X and Y are solids, Y has the larger specific heat capacity.
D	At time T, X gained more internal energy than Y.

Ans: C

X has higher melting point, not boiling point – A is wrong

Y takes a longer time to melt, hence more heat needs to be supplied to Y to melt. Since the masses of X and Y are the same, Y has the larger specific latent heat of fusion. B is wrong

Since the heating rate is constant and equal for both masses, the heat supplied is proportional to the time elapsed in heating. Hence, the inverse of the gradient corresponds to the specific heat capacity. (gradient = $\frac{\text{change in temperature}}{\text{change in time}}$,

inverse of gradient = $\frac{\text{change in time}}{\text{change in temperature}} \propto \frac{\text{heat supplied}}{\text{change in temperature}}$)

Hence the smaller the gradient, the larger the heat capacity. C is correct

There is not enough information to determine D.

16	A ball is held between two fixed points A and B by means of two stretched springs, as shown